CYBERVAULT-CTF

Integration, Testing & Documentation

IE3132 – Penetration Testing
Assignment 2 – CTF Play Box

Individual Contribution: Member 4

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GitHub Repository:
https://github.com/kkjanura123/CyberVault-CTF.git

1. Integration Testing

The CTFd platform was used to host and manage the CyberVault-CTF challenges. Integration testing focuses on checking challenge availability and verifying that the flag-submission mechanism behaves as expected.

The checks include accessing the challenges, submitting correct and incorrect flags, and identifying any integration issues.

2. Stage 1 Functional Testing

Stage 1, “The Teller's Photo,” is a steganography challenge. I inspected the image metadata using ExifTool and identified the clue cashier42. I then used Steghide to extract the hidden file, secret.txt.

The extracted file should be inspected to confirm the flag. The correct and incorrect flag-submission tests should also be completed in CTFd and their actual results recorded.

3. Test Results

Test	Expected Result	Result
Image metadata inspection	Identify the clue	Pass
Hidden-file extraction	Extract secret.txt	Pass
Reset and checksum check	Original and challenge images match after reset	Pass
Correct flag submission	Flag accepted	Pending Verification
Incorrect flag submission	Flag rejected	Pending Verification

4. Reset Testing and Improvements

I executed the Stage 1 reset script and compared the SHA-256 checksums of the original and challenge images. The checksums matched after the reset operation.

An improvement is to make the script check whether the copy operation succeeds before displaying a success message. The reset procedure should also define how temporary extracted files are handled.

5. Conclusion

Stage 1 metadata inspection and hidden-file extraction were completed successfully. The reset script was executed, and the image checksums matched afterward. The remaining CTFd flag-submission tests should be verified and documented before the final demonstration.